CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use graph paper for Q. 16.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The lines $x - 2y = 0$ and $3x + 4y - 20 = 0$ are: 1

(a) intersecting **(b)** parallel **(c)** coincident **(d)** none of these

2. The number of solutions of $2x + 3y = 9$ and $4x + 6y = 18$ is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** infinitely many

3. The value of k for which $x + 2y = 3$ and $5x + ky + 7 = 0$ has no solution is: 1

(a) 10 **(b)** 5 **(c)** −10 **(d)** 2

4. If $x = a,\ y = b$ is the solution of $x - y = 2$ and $x + y = 4$, then (a, b) is: 1

(a) $(3, 1)$ **(b)** $(1, 3)$ **(c)** $(2, 2)$ **(d)** $(4, 0)$

5. A student states: “Two lines that never intersect represent a consistent pair.” The correct response is: 1

(a) Correct

(b) Incorrect — parallel lines give no solution, so the pair is inconsistent

(c) Correct only for $x = 0$

(d) Cannot be determined

6. The lines $x = 2$ and $y = 3$ intersect at: 1

(a) $(3, 2)$ **(b)** $(2, 3)$ **(c)** $(0, 0)$ **(d)** $(2, 0)$

7. For $x + y = 7$ and $2x - y = 2$, Student A gets $(3, 4)$ and Student B gets $(4, 3)$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Neither is correct

8. If the lines $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then k is: 1

(a) $\frac{15}{4}$ **(b)** $\frac{4}{15}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{3}{2}$

9. The pair of equations $y = 0$ and $y = -7$ has: 1

(a) one solution **(b)** two solutions

(c) infinitely many solutions **(d)** no solution

10. One number is twice another and their sum is 36. The numbers are: 1

(a) $24, 12$ **(b)** $20, 16$ **(c)** $18, 18$ **(d)** $26, 10$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by the elimination method: $x + y = 5$ and $2x - 3y = 4$. 2

12. Check whether the pair $5x - 4y + 8 = 0$ and $7x + 6y - 9 = 0$ is consistent. 2

13. Find a and b for which the pair $2x + 3y = 7$ and $(a - b)x + (a + b)y = 3a + b - 2$ has infinitely many solutions. 2

14. The difference between two numbers is 26 and one number is three times the other. Find them. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by the substitution method: $0.2x + 0.3y = 1.3$ and $0.4x + 0.5y = 2.3$. 3

16. Draw the graphs of $2x + y = 6$ and $2x - y + 2 = 0$. Shade the region bounded by these lines and the x-axis, and find its vertices. 3

17. A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid ₹27 for a book kept for seven days, while Susy paid ₹21 for a book kept for five days. Find the fixed charge and the charge for each extra day. 3

OR

Two years ago, a father was five times as old as his son. Two years later, his age will be 8 more than three times the age of the son. Find their present ages. 3

18. Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If they travel in the same direction, they meet in 5 hours; if they travel towards each other, they meet in 1 hour. Find the speeds of the two cars. 3

OR

Of two supplementary angles, the larger exceeds the smaller by 18°. Find the angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** In a competitive test, 3 marks are awarded for every correct answer and 1 mark is deducted for every wrong answer. Yash scored 40 marks. Had 4 marks been awarded for each correct answer and 2 marks deducted for each wrong answer, he would have scored 50 marks. 5

(i) Form the pair of linear equations.

(ii) Solve them.

(iii) How many questions were there in the test (all attempted)?

OR

A shopkeeper sells a saree at 8% profit and a sweater at 10% discount, thereby getting ₹1008. If she had sold the saree at 10% profit and the sweater at 8% discount, she would have got ₹1028. Find the cost price of the saree and the list price of the sweater. 5

20. The area of a rectangle gets reduced by 9 square units if its length is reduced by 5 units and breadth is increased by 3 units. If the length is increased by 3 units and breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle and verify. 5

OR

A train covered a certain distance at a uniform speed. If the train had been 10 km/h faster, it would have taken 2 hours less; and if it had been 10 km/h slower, it would have taken 3 hours more. Find the distance covered. 5

END OF QUESTION PAPER — **Total: 40 Marks**